Lab 7: Fuzzing for Input Validation Bugs

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Assigned repo	SETAPESU26/L6_L7_35_customs_duty (lab7)
Application script	customs_duty.py (the handout's processor.py)
Test script	test_customs_duty.py (the handout's test_processor.py)

Step 1: Install dependencies

```
pip install hypothesis
pip install pytest
```

Step 2: Run the buggy application

Command: `python customs_duty.py`. A valid run works, but a declaration without a ':' crashes the program with a ValueError.

```
PS D:\chrome downloads\SE_LAB\LAB7> python customs_duty.py
==== Running parse_declaration ====
Enter category:value (e.g. electronics:25000): electronics:25000
Category and Value: ('electronics', 25000.0)

==== Running duty_amount ====
Enter value@rate (e.g. 25000@20): 25000@20
Duty Amount: 5000.0

==== Running hs_chapter ====
Enter an HS code (e.g. 8517.12): 8517.12
Chapter: 85
PS D:\chrome downloads\SE_LAB\LAB7> python customs_duty.py
==== Running parse_declaration ====
Enter category:value (e.g. electronics:25000): electronics
Traceback (most recent call last):
  File "D:\chrome downloads\SE_LAB\LAB7\customs_duty.py", line 29, in <module>
    main()
    ~~~~^^
  File "D:\chrome downloads\SE_LAB\LAB7\customs_duty.py", line 15, in main
    result = parse_declaration(user_input)
  File "D:\chrome downloads\SE_LAB\LAB7\customs_duty.py", line 2, in parse_declaration
    category, value = text.split(':')
    ^^^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
PS D:\chrome downloads\SE_LAB\LAB7>
```

Step 3: Hypothesis tests written in test_customs_duty.py

Test	Strategy	Property checked
test_parse_declaration_no_crash	st.text() | st.none()	never raises an exception
test_duty_amount_no_crash	st.text() | st.none()	never raises an exception
test_hs_chapter_no_crash	st.text() | st.none()	never raises an exception
test_parse_declaration_any_parts_no_crash	two st.text() joined by ':'	never raises an exception

Test	Strategy	Property checked
test_duty_amount_is_never_negative_or_nan	st.floats() joined by '@', or st.text()	result is None or a finite number >= 0
test_hs_chapter_is_a_real_chapter	st.text()	result is None or a chapter from 1 to 97
test_parse_declaration_valid	lowercase text + st.floats(0..1e9)	valid input returns (category, value)
test_duty_amount_valid	st.floats(0..1e9) and st.floats(0..100)	valid input returns value * rate / 100
test_hs_chapter_valid	st.integers() built into 'CCHH.SS'	valid code returns its chapter

Step 4: Run the tests against the buggy application (failing)

Command: `python -m pytest test_customs_duty.py`. Result: **6 failed, 3 passed**. The failing inputs Hypothesis found:

Function	Input found by Hypothesis	Bug
parse_declaration	None	AttributeError: 'NoneType' object has no attribute 'split'
parse_declaration	'' (no ':')	ValueError: not enough values to unpack
parse_declaration	':' (empty value)	ValueError: could not convert string to float: ''
parse_declaration	text with two ':'	ValueError: too many values to unpack
duty_amount	None / no '@' / '@'	AttributeError and ValueError crashes, as above
duty_amount	'1.0@-1.0'	returns a negative duty (-0.01); 'nan' and 'inf' were also accepted
hs_chapter	None / ''	AttributeError / ValueError: invalid literal for int()
hs_chapter	'0'	returns chapter 0, which does not exist (also accepted '-1')

Full pytest output (continued over the following screenshots):

Failing test output, part 1

```
PS D:\chrome downloads\SE_LAB\LAB7> python -m pytest test_customs_duty.py
============================================ test session starts ============================================
platform win32 -- Python 3.13.13, pytest-9.1.1, pluggy-1.7.0
rootdir: D:\chrome downloads\SE_LAB\LAB7
plugins: anyio-4.14.2, hypothesis-6.170.15, platformdirs-4.14.5, typeguard-4.6.0
collected 9 items

test_customs_duty.py FFFFFF...                                                                      [100%]

================================================ FAILURES ================================================
_____________________________________ test_parse_declaration_no_crash _____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 17, in test_parse_declaration_no_crash
  |     def test_parse_declaration_no_crash(s):
  |                    ^^^
  |   File "C:\Users\SAI HEMANTH\AppData\Local\Programs\Python\Python313\Lib\site-packages\hypothesis\core.py", line
2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 19, in test_parse_declaration_no_crash
    |     parse_declaration(s)
    |     ~~~~~~~~~~~~~~~~~~^^^
    |   File "D:\chrome downloads\SE_LAB\LAB7\customs_duty.py", line 3, in parse_declaration
    |     return category.strip().lower(), float(value)
    |                                      ~~~~~^^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 21, in test_parse_declaration_no_crash
    |     assert False, f"parse_declaration crashed with: {e}"
    | AssertionError: parse_declaration crashed with: could not convert string to float: ''
    | assert False
    | Failing test case: test_parse_declaration_no_crash(
    |     s=':',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 19, in test_parse_declaration_no_crash
    |     parse_declaration(s)
    |     ~~~~~~~~~~~~~~~~~~^^^
    |   File "D:\chrome downloads\SE_LAB\LAB7\customs_duty.py", line 2, in parse_declaration
    |     category, value = text.split(':')
    |     ^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 21, in test_parse_declaration_no_crash
    |     assert False, f"parse_declaration crashed with: {e}"
    | AssertionError: parse_declaration crashed with: not enough values to unpack (expected 2, got 1)
```

Failing test output, part 2

```
    | assert False
    | Failing test case: test_parse_declaration_no_crash(
    |     s='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 19, in test_parse_declaration_no_crash
    |     parse_declaration(s)
    |     ~~~~~~~~~~~~~~~~~~^^^
    |   File "D:\chrome downloads\SE_LAB\LAB7\customs_duty.py", line 2, in parse_declaration
    |     category, value = text.split(':')
    |                       ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 21, in test_parse_declaration_no_crash
    |     assert False, f"parse_declaration crashed with: {e}"
    | AssertionError: parse_declaration crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_parse_declaration_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____________________________________ test_duty_amount_no_crash _____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 25, in test_duty_amount_no_crash
  |     def test_duty_amount_no_crash(s):
  |                    ^^^
  |   File "C:\Users\SAI HEMANTH\AppData\Local\Programs\Python\Python313\Lib\site-packages\hypothesis\core.py", line
2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 27, in test_duty_amount_no_crash
    |     duty_amount(s)
    |     ~~~~~~~~~~~~^^^
    |   File "D:\chrome downloads\SE_LAB\LAB7\customs_duty.py", line 7, in duty_amount
    |     return float(value) * float(rate) / 100
    |            ~~~~~^^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 29, in test_duty_amount_no_crash
    |     assert False, f"duty_amount crashed with: {e}"
    | AssertionError: duty_amount crashed with: could not convert string to float: ''
    | assert False
    | Failing test case: test_duty_amount_no_crash(
    |     s='@',
    | )
    +---------------- 2 ----------------
```

Failing test output, part 3

```
    | Traceback (most recent call last):
    |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 27, in test_duty_amount_no_crash
    |     duty_amount(s)
    |     ~~~~~~~~~~~~^^^
    |   File "D:\chrome downloads\SE_LAB\LAB7\customs_duty.py", line 6, in duty_amount
    |     value, rate = text.split('@')
    |     ^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 29, in test_duty_amount_no_crash
    |     assert False, f"duty_amount crashed with: {e}"
    | AssertionError: duty_amount crashed with: not enough values to unpack (expected 2, got 1)
    | assert False
    | Failing test case: test_duty_amount_no_crash(
    |     s='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 27, in test_duty_amount_no_crash
    |     duty_amount(s)
    |     ~~~~~~~~~~~~^^^
    |   File "D:\chrome downloads\SE_LAB\LAB7\customs_duty.py", line 6, in duty_amount
    |     value, rate = text.split('@')
    |                   ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 29, in test_duty_amount_no_crash
    |     assert False, f"duty_amount crashed with: {e}"
    | AssertionError: duty_amount crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_duty_amount_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____________________________________ test_hs_chapter_no_crash _____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 33, in test_hs_chapter_no_crash
  |     def test_hs_chapter_no_crash(s):
  |                    ^^^
  |   File "C:\Users\SAI HEMANTH\AppData\Local\Programs\Python\Python313\Lib\site-packages\hypothesis\core.py", line
2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 35, in test_hs_chapter_no_crash
    |     hs_chapter(s)
    |     ~~~~~~~~~~~^^^
    |   File "D:\chrome downloads\SE_LAB\LAB7\customs_duty.py", line 10, in hs_chapter
```

Failing test output, part 4

```
    |     return int(code.split('.')[0][:2])
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 37, in test_hs_chapter_no_crash
    |     assert False, f"hs_chapter crashed with: {e}"
    | AssertionError: hs_chapter crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_hs_chapter_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 35, in test_hs_chapter_no_crash
    |     hs_chapter(s)
    |     ~~~~~~~~~~~^^^
    |   File "D:\chrome downloads\SE_LAB\LAB7\customs_duty.py", line 10, in hs_chapter
    |     return int(code.split('.')[0][:2])
    |                ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 37, in test_hs_chapter_no_crash
    |     assert False, f"hs_chapter crashed with: {e}"
    | AssertionError: hs_chapter crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_hs_chapter_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____________________________ test_parse_declaration_any_parts_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 43, in test_parse_declaration_any_parts_no_crash
  |     def test_parse_declaration_any_parts_no_crash(category, value):
  |                    ^^^
  |   File "C:\Users\SAI HEMANTH\AppData\Local\Programs\Python\Python313\Lib\site-packages\hypothesis\core.py", line
2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 45, in
test_parse_declaration_any_parts_no_crash
    |     parse_declaration(f"{category}:{value}")
    |     ~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^
    |   File "D:\chrome downloads\SE_LAB\LAB7\customs_duty.py", line 2, in parse_declaration
    |     category, value = text.split(':')
    |     ^^^^^^^^^^^^^^
    | ValueError: too many values to unpack (expected 2)
    |
    | During handling of the above exception, another exception occurred:
```

Failing test output, part 5

```
    |
    | Traceback (most recent call last):
    |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 47, in
test_parse_declaration_any_parts_no_crash
    |     assert False, f"parse_declaration crashed with: {e}"
    | AssertionError: parse_declaration crashed with: too many values to unpack (expected 2)
    | assert False
    | Failing test case: test_parse_declaration_any_parts_no_crash(
    |     category='',
    |     value=':',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 45, in
test_parse_declaration_any_parts_no_crash
    |     parse_declaration(f"{category}:{value}")
    |     ~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^
    |   File "D:\chrome downloads\SE_LAB\LAB7\customs_duty.py", line 3, in parse_declaration
    |     return category.strip().lower(), float(value)
    |                                      ~~~~~^^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\chrome downloads\SE_LAB\LAB7\test_customs_duty.py", line 47, in
test_parse_declaration_any_parts_no_crash
    |     assert False, f"parse_declaration crashed with: {e}"
    | AssertionError: parse_declaration crashed with: could not convert string to float: ''
    | assert False
    | Failing test case: test_parse_declaration_any_parts_no_crash(
    |     category='',
    |     value='',
    | )
    +-----------------------------------
_____________________________ test_duty_amount_is_never_negative_or_nan _____________________________

    @given(st.builds("{}@{}".format, st.floats(), st.floats()) | st.text())
>   def test_duty_amount_is_never_negative_or_nan(s):
                ^^^

test_customs_duty.py:51:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

s = '1.0@-1.0'

    @given(st.builds("{}@{}".format, st.floats(), st.floats()) | st.text())
    def test_duty_amount_is_never_negative_or_nan(s):
        try:
            result = duty_amount(s)
        except Exception:
            return  # crashes are reported by test_duty_amount_no_crash
>       assert result is None or (math.isfinite(result) and result >= 0)
E       assert (-0.01 is None or (True and -0.01 >= 0))
E        +  where True = <built-in function isfinite>(-0.01)
```

Failing test output, part 6

```
E        +    where <built-in function isfinite> = math.isfinite
E       Failing test case: test_duty_amount_is_never_negative_or_nan(
E           s='1.0@-1.0',
E       )

test_customs_duty.py:56: AssertionError
_____________________________________ test_hs_chapter_is_a_real_chapter _____________________________________

    @given(st.text())
>   def test_hs_chapter_is_a_real_chapter(s):
                  ^^^

test_customs_duty.py:60:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

s = '0'

    @given(st.text())
    def test_hs_chapter_is_a_real_chapter(s):
        try:
            result = hs_chapter(s)
        except Exception:
            return  # crashes are reported by test_hs_chapter_no_crash
>       assert result is None or 1 <= result <= 97
E       assert (0 is None or 1 <= 0)
E       Failing test case: test_hs_chapter_is_a_real_chapter(
E           s='0',
E       )

test_customs_duty.py:65: AssertionError
========================================== short test summary info ==========================================
FAILED test_customs_duty.py::test_parse_declaration_no_crash - ExceptionGroup: Hypothesis found 3 distinct ...
FAILED test_customs_duty.py::test_duty_amount_no_crash - ExceptionGroup: Hypothesis found 3 distinct failur...
FAILED test_customs_duty.py::test_hs_chapter_no_crash - ExceptionGroup: Hypothesis found 2 distinct failure...
FAILED test_customs_duty.py::test_parse_declaration_any_parts_no_crash - ExceptionGroup: Hypothesis found 2...
FAILED test_customs_duty.py::test_duty_amount_is_never_negative_or_nan - assert (-0.01 is None or (True and...
FAILED test_customs_duty.py::test_hs_chapter_is_a_real_chapter - assert (0 is None or 1 <= 0)
========================================== 6 failed, 3 passed in 1.77s ==========================================
PS D:\chrome downloads\SE_LAB\LAB7>
```

Step 5: Fix the application and run it again

Function	Fix
all three	Return None for anything that is not a string, instead of crashing
parse_declaration	Require exactly one ':', a non-empty category, and a finite value >= 0
duty_amount	Require exactly one '@', a finite value >= 0 and a rate from 0 to 100; reject 'nan', 'inf' and overflow; compute value * (rate / 100) so it cannot overflow
hs_chapter	Require at least 2 ASCII digits before the '.', and a chapter from 01 to 97
main()	Print a clear 'Invalid ...' message instead of a traceback

Command: `python customs_duty.py`. Valid input gives the same results as before, and invalid input now prints a message instead of crashing.

```
PS D:\chrome downloads\SE_LAB\LAB7> python customs_duty.py
==== Running parse_declaration ====
Enter category:value (e.g. electronics:25000): electronics:25000
Category and Value: ('electronics', 25000.0)

==== Running duty_amount ====
Enter value@rate (e.g. 25000@20): 25000@20
Duty Amount: 5000.0

==== Running hs_chapter ====
Enter an HS code (e.g. 8517.12): 8517.12
Chapter: 85
PS D:\chrome downloads\SE_LAB\LAB7> python customs_duty.py
==== Running parse_declaration ====
Enter category:value (e.g. electronics:25000): electronics
Invalid declaration: use category:value with a non-negative number

==== Running duty_amount ====
Enter value@rate (e.g. 25000@20): 25000@-5
Invalid input: use value@rate with value >= 0 and rate between 0 and 100

==== Running hs_chapter ====
Enter an HS code (e.g. 8517.12): abc
Invalid HS code: it must start with a 2-digit chapter from 01 to 97
PS D:\chrome downloads\SE_LAB\LAB7>
```

Step 6: Re-run the tests against the fixed application (passing)

Command: `python -m pytest test_customs_duty.py -v`. Result: **all 9 tests passed**.

```
PS D:\chrome downloads\SE_LAB\LAB7> python -m pytest test_customs_duty.py -v
========================================== test session starts ==========================================
platform win32 -- Python 3.13.13, pytest-9.1.1, pluggy-1.7.0 -- C:\Users\SAI
HEMANTH\AppData\Local\Programs\Python\Python313\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: D:\chrome downloads\SE_LAB\LAB7
plugins: anyio-4.14.2, hypothesis-6.170.15, platformdirs-4.14.5, typeguard-4.6.0
collecting ... collected 9 items

test_customs_duty.py::test_parse_declaration_no_crash PASSED                                    [ 11%]
test_customs_duty.py::test_duty_amount_no_crash PASSED                                          [ 22%]
test_customs_duty.py::test_hs_chapter_no_crash PASSED                                           [ 33%]
test_customs_duty.py::test_parse_declaration_any_parts_no_crash PASSED                          [ 44%]
test_customs_duty.py::test_duty_amount_is_never_negative_or_nan PASSED                          [ 55%]
test_customs_duty.py::test_hs_chapter_is_a_real_chapter PASSED                                  [ 66%]
test_customs_duty.py::test_parse_declaration_valid PASSED                                       [ 77%]
test_customs_duty.py::test_duty_amount_valid PASSED                                             [ 88%]
test_customs_duty.py::test_hs_chapter_valid PASSED                                              [100%]

========================================== 9 passed in 1.50s ==========================================
PS D:\chrome downloads\SE_LAB\LAB7>
```